Various applications of Lidar

Autonomous vehicles. Used for sensing the surroundings, making it possible for driverless vehicles to avoid collisions and navigate safely.

Mapping and surveying. Lidar is used extensively in cartography and surveying applications to create detailed 3D maps of terrain, buildings, and other features.

Agriculture. Lidar is used in precision agriculture to optimise crop yield and reduce waste.

Forestry. Lidar is used in forestry applications to analyse the volume of trees and monitor tree growth and forest health, as well as to identify areas of high risk for forest fires and other hazards.

Infrastructure inspection. Lidar is used to detect small defects in the surface of bridges, dams, and pipelines.

Archaeology. Lidar is used in archaeology to create detailed 3D models of historical sites and artefacts, as it can help uncover hidden features and structures that may not be visible to the naked eye.

Security and surveillance. Lidar is used in security and surveillance applications to detect and track moving objects in real time. It can also be used to create 3D models of indoor and outdoor environments, allowing security personnel to identify potential threats and vulnerabilities.

moving parts, which makes them more reliable and durable. Moreover, solid-state lidars are typically smaller and lighter, and have higher efficiency. They also require lesser energy to work. Compared to mechanical spinning lidars, solid-state lidars are cheaper to manufacture.

But one of the main drawbacks of solid-state lidars is that they have narrower field of view. Some examples of solid-state lidars are Velarray M1600, Opsys SP3C, and LeddarSteer.

Wider field of view

The lidar manufacturers are now opting for lidar modules with a wider field of view. The wider field of view is advantageous for many applications as it allows more comprehensive and detailed scanning of the environment. It offers better coverage in a single scan, which can reduce the time required to scan an entire environment and provide more comprehensive coverage.

Moreover, the wider field of view allows for better object detection. It also offers better accuracy and resolution of 3D maps generated by lidar systems, which can be useful for applications such as autonomous vehicles, robotics, and industrial automation. For vehicles, a larger field of view can minimise blind spots that may be present with narrower sensors, which can increase situational awareness and improve safety.

However, a wider field of view may come with tradeoffs in terms of resolution, accuracy, and cost, depending on the specific lidar system and application requirements. Some lidars with a wider field of view are Velodyne Alpha prime, ML-30s+, Vista-X120, XenomatiX, Velarray M1600, OSDome, and Aeries II.

Faster object detection

Lidars are getting faster with the development of new technologies and advancements in hardware and software. Lidar sensors like XenomatiX, ML-30s+, Rev 7, Velarray M1600, Opsys, and LeddarSteer provide faster data output. A faster lidar sensor can capture more data points in a shorter amount of time, which can be crucial for real-time applications, such as autonomous driving.

The development of solid-state lidar technology is one of the key drivers of the increased speed of lidar systems. Another factor driving the faster development of lidar sensors is the more powerful and efficient data processing algorithms. As lidar sensors capture larger amounts of data, more advanced software algorithms are needed to process and analyse this data in real time, allowing the system to make decisions and take actions quickly and accurately.

Improvement in accuracy

Lidar sensors are becoming more accurate in terms of their ability to measure the distance between the sensor and surrounding objects, as well as their ability to capture more detailed information about the environment. Some lidars with improved efficiency are Ouster REV7 lidar, LeddarSteer, and ML-30s+.

One of the key factors driving the improvement in lidar accuracy is the development of higher-resolution sensors. Lidar sensors with higher resolution can capture more data points per scan, which allows for more detailed and accurate 3D maps of the environment. Furthermore, the improvement of advanced algorithms for processing lidar data, machine learning, and other advanced techniques to analyse lidar data are one of the main reasons for the improved accuracy of the lidars.

The post-processing, which corrects errors caused by factors such as sensor noise, environmental interference, and other sources of noise, have also improved and resulted in higher accuracy of lidar sensor.

Improvement in dark object detection

The newer lidar systems are more capable of working in dark environments than the older systems. One reason for this is the development of technologies that improve the sensitivity of lidar sensors to low levels of light. Ouster REV7 lidar sensor employs back-side illumination technology to improve night imaging. Lidars such as Cepton Vista-X120 use sophisticated signal processing techniques to filter out the noise and improve the quality of data captured